



Neutrinos via Charm Decays in Astrophysical Sources

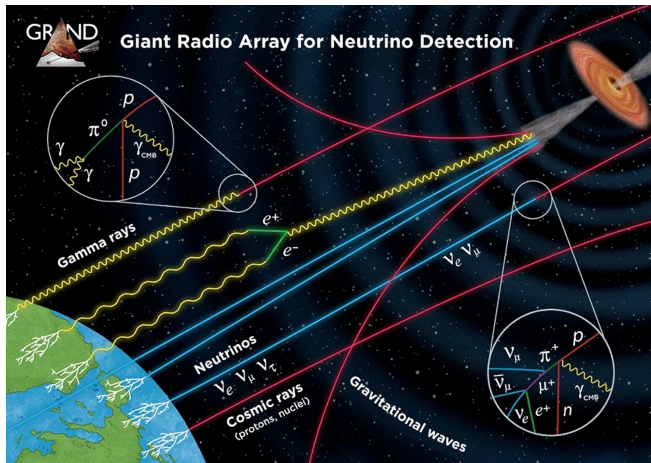
TALK IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE

Fritz Ali Agildere

September 3, 2024

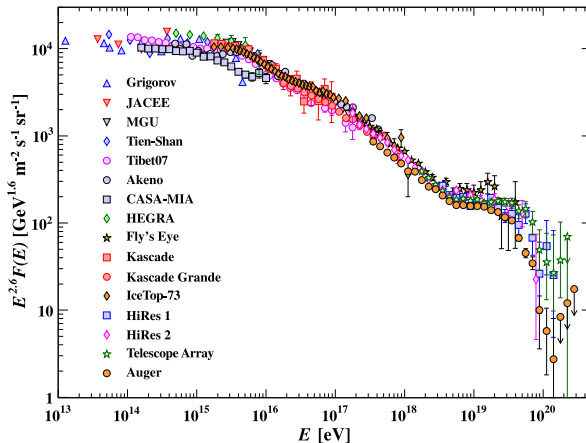
Technical University of Dortmund & Ruhr University Bochum

Multimessenger Astrophysics



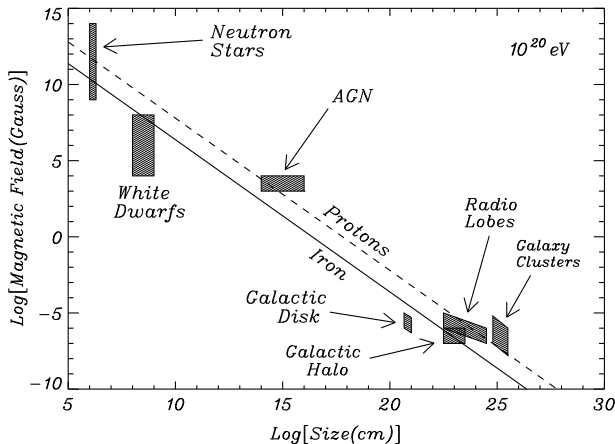
X. L. Qian et al.
Nucl. Sci. Tech. (2021)

Cosmic Ray Spectrum & Cutoff



K. A. Olive et al.
Chin. Phys. C (2014)

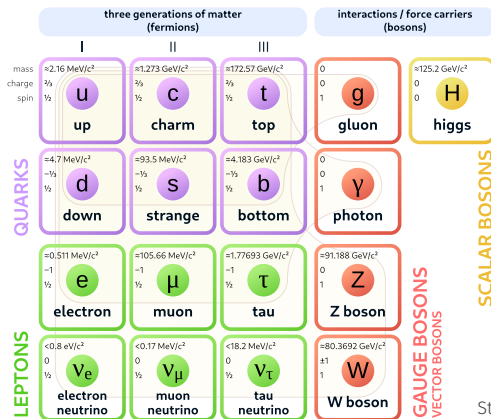
Plausible Cosmic Ray Origins



A. V. Olinto
AIP Conf. Proc. (2000)

Elementary Particles & Hadrons

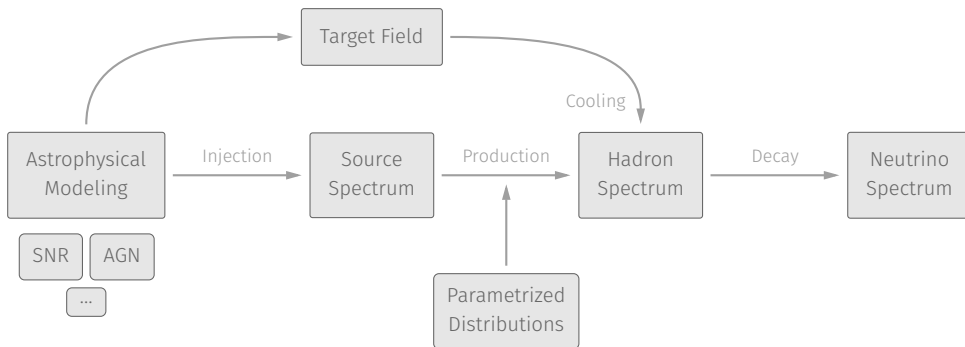
Standard Model of Elementary Particles

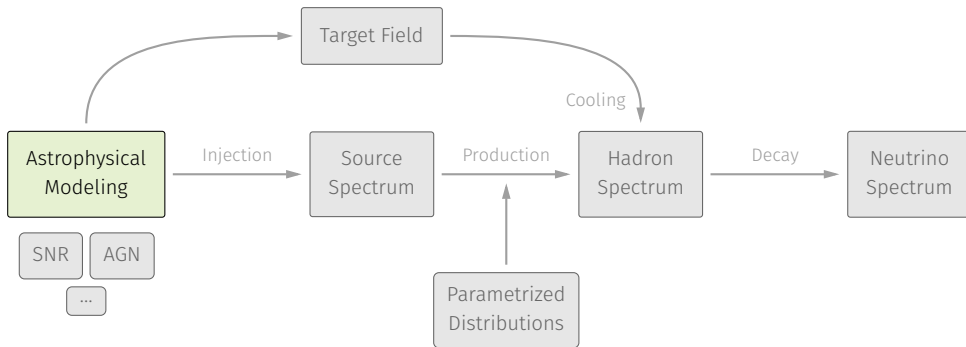


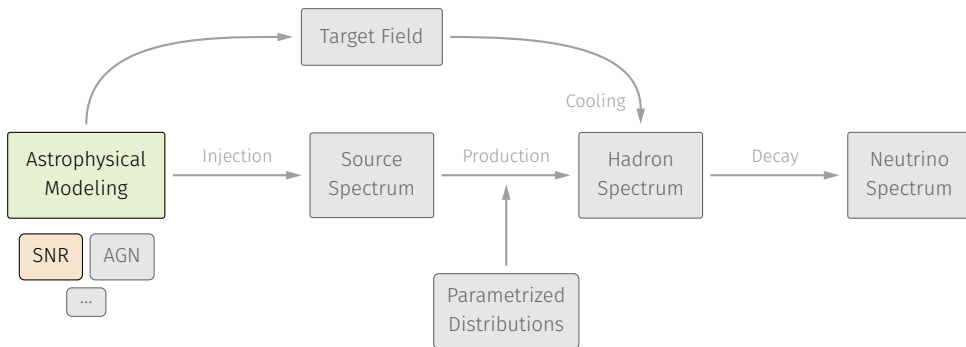
Standard Model
Wikimedia Commons (2019)

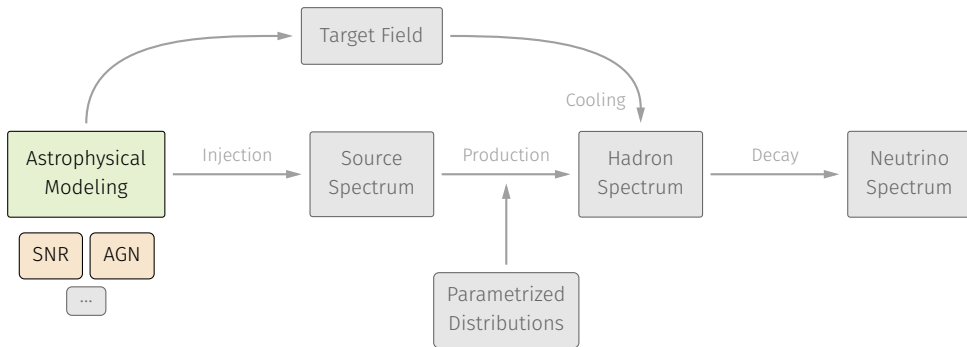
Relevant Hadron Properties

h	q_h	τ_h / s	m_h / GeV	s_h / GeV^2	$B_{h \rightarrow \nu}$
p	uud	∞	0.94	15.98	0.000
π^+	$u\bar{d}$	26.0×10^{-9}	0.14	10.23	1.000
K^+	$u\bar{s}$	12.4×10^{-9}	0.49	12.62	0.636
D_0	$c\bar{u}$	0.4×10^{-12}	1.86	0.45	0.067
D^+	$c\bar{d}$	1.0×10^{-12}	1.87	0.40	0.176
D_s^+	$c\bar{s}$	0.5×10^{-12}	1.97	0.71	0.065
Λ_c^+	udc	0.2×10^{-12}	2.29	1.61	0.045

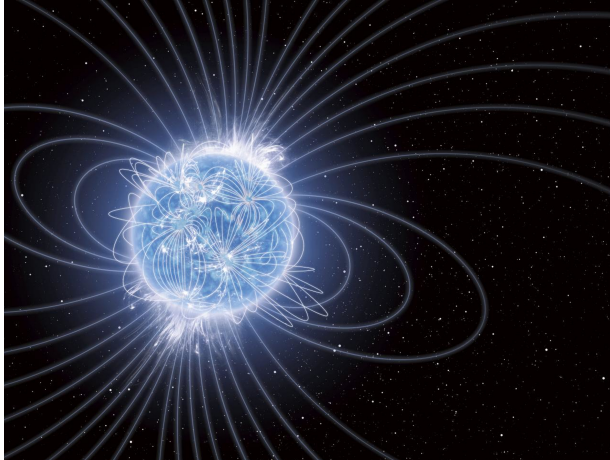






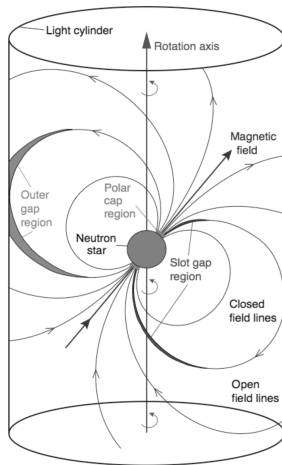


Young Neutron Stars



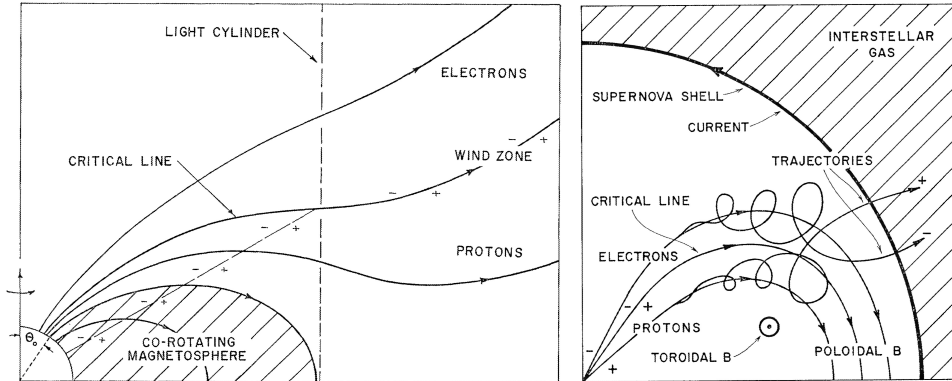
R. Takeshige
Magnetar (2020)

Dipolar Pulsar Model



E. Aliu et al.
Science (2008)

Particle Acceleration



P. Goldreich, W. H. Julian *ApJ* (1969)



Magnetar Spindown

- Electromagnetic Luminosity

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

Magnetar Spindown

- Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

- Characteristic Timescale

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

Magnetar Spindown

- Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

- Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

- Monotonic Energy

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

■ Monotonic Energy

$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

■ Monotonic Energy

$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

■ Monotonic Energy

$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

Magnetar Spindown

■ Electromagnetic Luminosity

$$L = \frac{B_{ns}^2 R_{ns}^6 \omega_0^4 (1 + \sin^2 \chi)}{4c^3 (1 + t/t_{sd})^2}$$

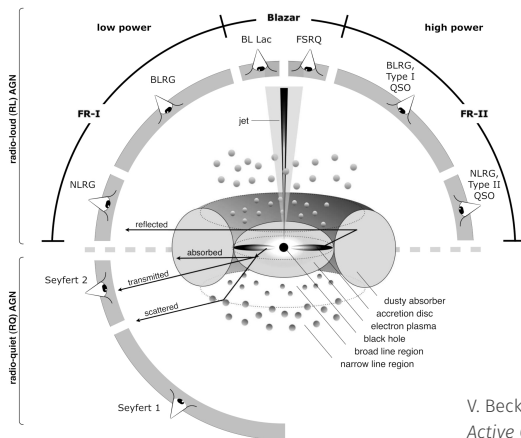
■ Characteristic Timescale

$$t_{sd} = \frac{2I_{ns} c^3}{B_{ns}^2 R_{ns}^6 \omega_0^2 (1 + \sin^2 \chi)}$$

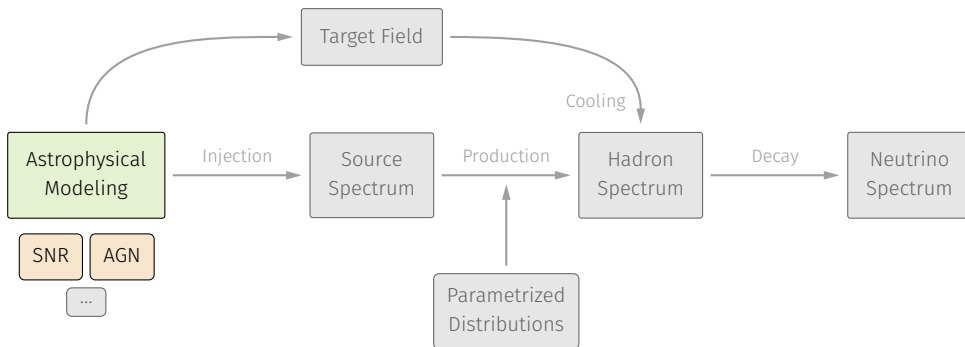
■ Monotonic Energy

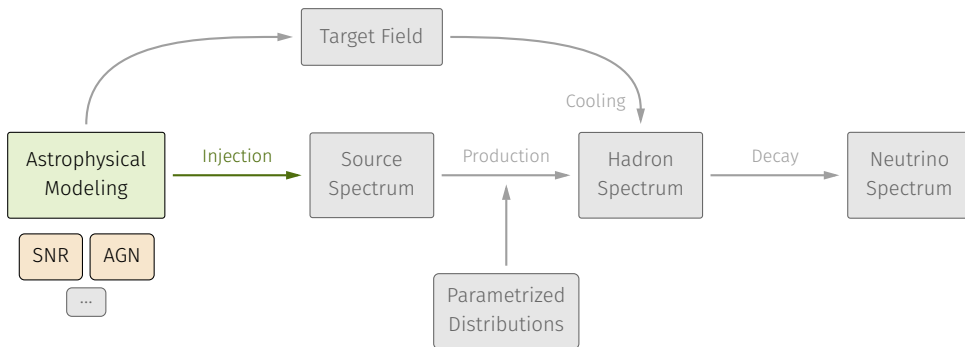
$$E^M = \frac{\eta e B_{ns} R_{ns}^3 \omega_0^2}{2c^2 (1 + t/t_{sd})}$$

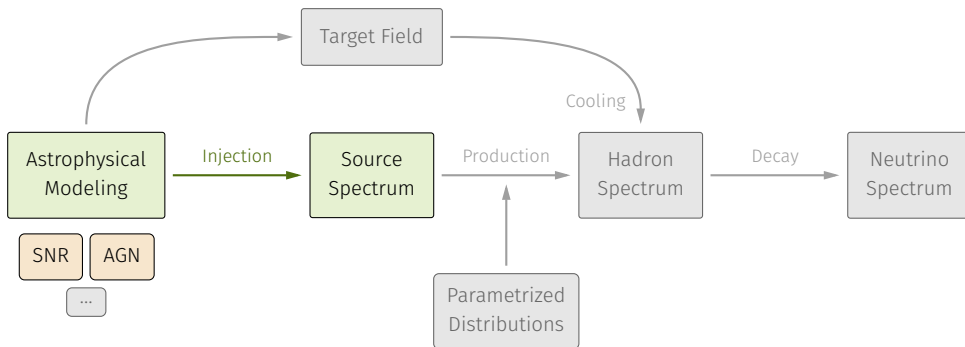
Active Galactic Nuclei



V. Beckmann, C. Shrader
Active Galactic Nuclei (2012)









Injected Primary Spectrum

- Monochromatic Proton Energy

Injected Primary Spectrum

■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

Injected Primary Spectrum

■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

Injected Primary Spectrum

■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

Injected Primary Spectrum

■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

Injected Primary Spectrum

■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

$$\frac{dN_p}{dE_p} \propto E_p^{-1}$$

Injected Primary Spectrum

■ Monochromatic Proton Energy

$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

$$\frac{dN_p}{dE_p} \propto E_p^{-1}$$

■ Stochastic Acceleration Power Law

Injected Primary Spectrum

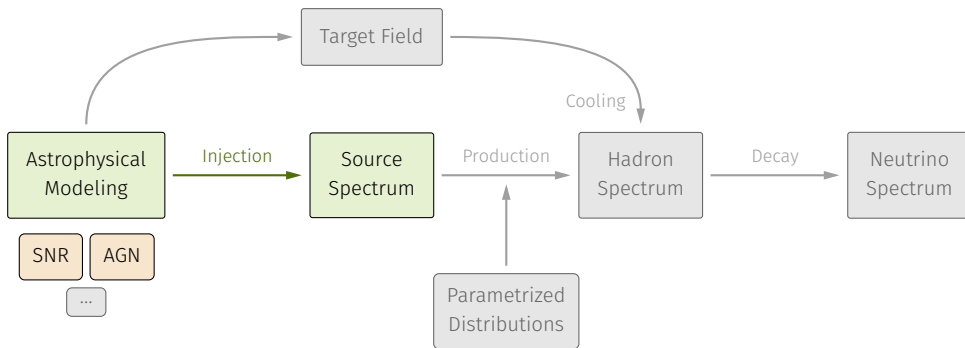
■ Monochromatic Proton Energy

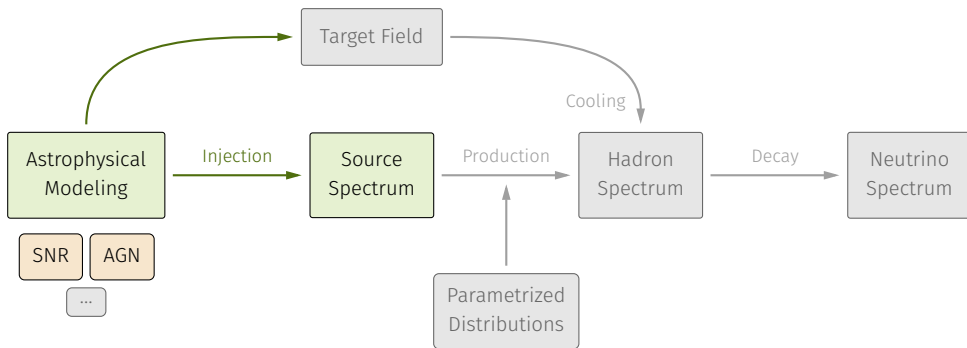
$$\frac{d\dot{N}_p}{dE_p} = \frac{B_{ns} R_{ns}^3 \omega_0^2}{ec(1 + t/t_{sd})} \delta(E_p - E^M)$$

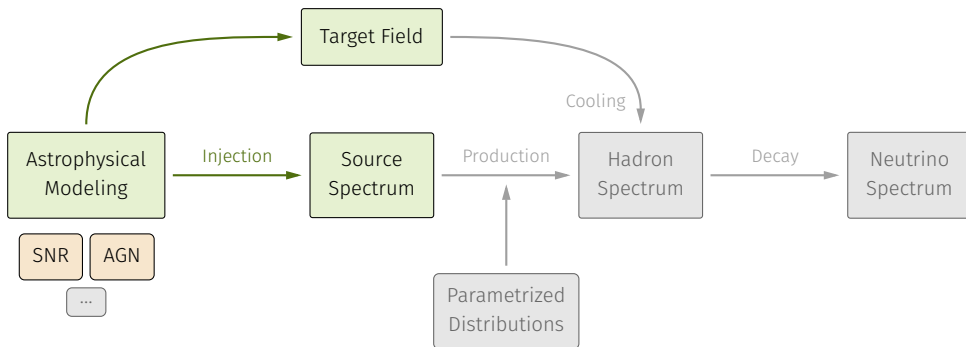
$$\frac{dN_p}{dE_p} \propto E_p^{-1}$$

■ Stochastic Acceleration Power Law

$$\frac{dN_p}{dE_p} \propto E_p^{-2}$$









Magnetar Environment

- Supernova Shell Radius

Magnetar Environment

- Supernova Shell Radius

$$r_{ej} = \beta_{ej} ct$$

Magnetar Environment

- Supernova Shell Radius

$$r_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

Magnetar Environment

- Supernova Shell Radius

$$r_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

Magnetar Environment

- Supernova Shell Radius

$$r_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

Magnetar Environment

- Supernova Shell Radius

$$r_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

Magnetar Environment

- Supernova Shell Radius

$$r_{ej} = \beta_{ej} ct$$

- Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

- Numerical Setup

Magnetar Environment

■ Supernova Shell Radius

$$r_{ej} = \beta_{ej} ct$$

■ Ejecta Number Density

$$n_{ej} = \frac{3M_{ej}}{4\pi R_{ej}^3 m_p}$$

■ Numerical Setup

$$\begin{array}{llll} I_{ns} = 10^{45} \text{ g cm}^2 & R_{ns} = 10^6 \text{ cm} & B_{ns} = 10^{14.5} \text{ G} & t_{sd} = 10^{3.5} \text{ s} \\ M_{ej} = 10 M_{\odot} & \beta_{ej} = 0.1 & \eta = 0.1 & \end{array}$$



Accretion Disk

- Thin Limit Scale Height

Accretion Disk

■ Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

Accretion Disk

- Thin Limit Scale Height

$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

- Parameter Choice

Accretion Disk

- Thin Limit Scale Height

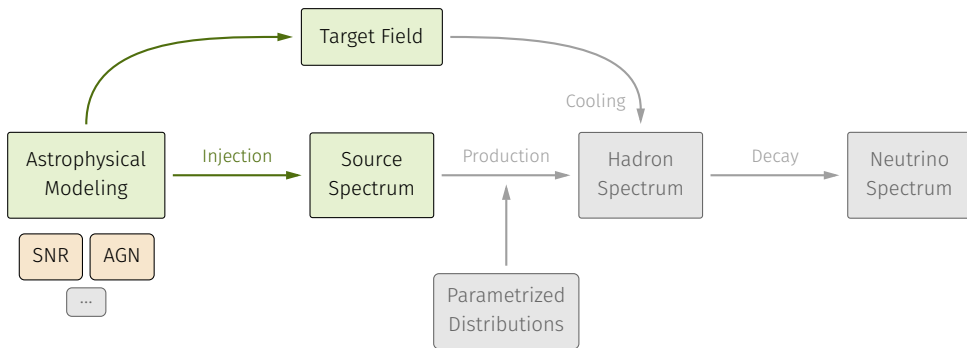
$$h^2 = \frac{u_{sn}^2 R_{dc}^3}{GM_{bh}}$$

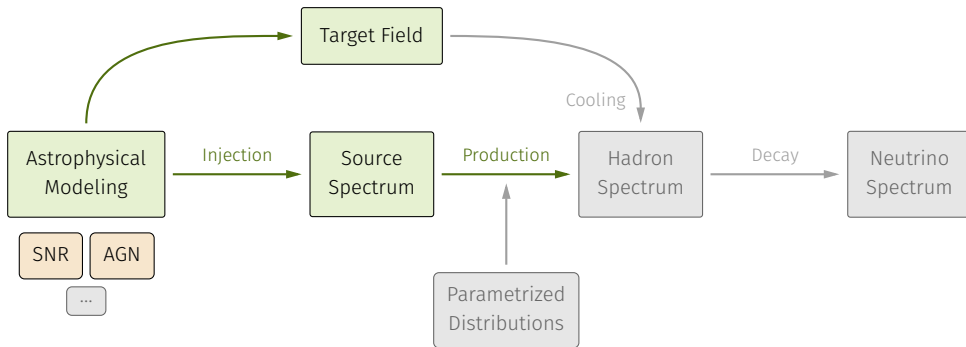
- Gaussian Density Profile

$$\rho \propto \exp\left(-\frac{GM_{bh} z^2}{2u_{sn}^2 R_{dc}^3}\right)$$

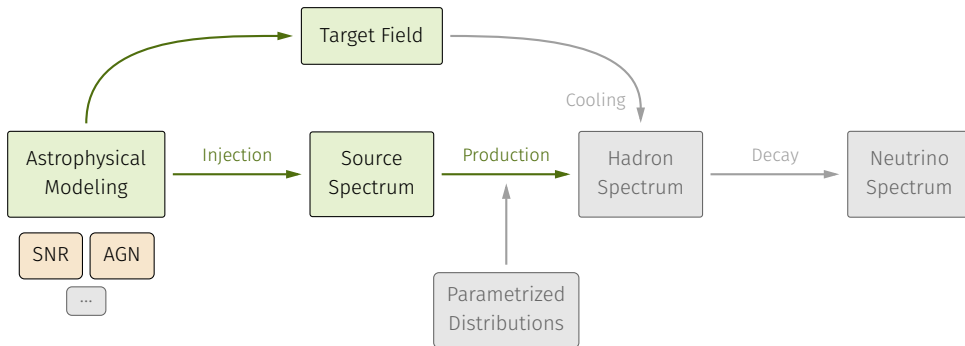
- Parameter Choice

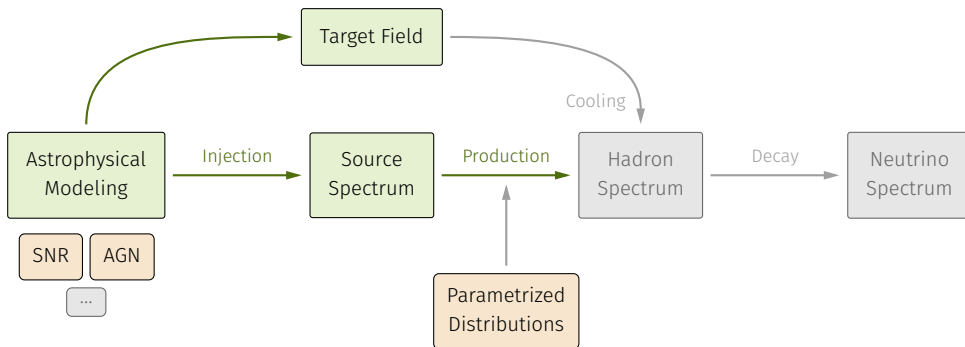
$$n_{dc} = 10^{14} \text{ cm}^{-3} \quad h = 10^{15} \text{ cm}$$

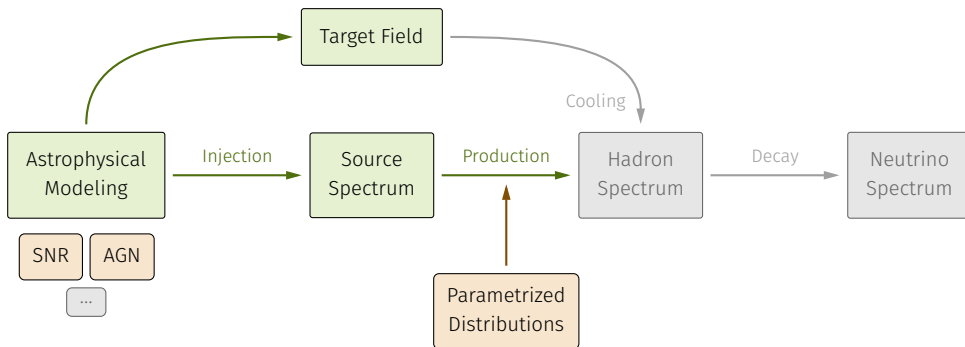




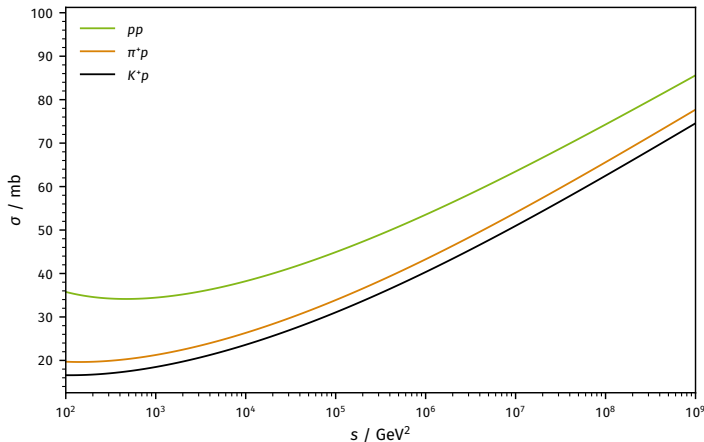








Hadron Inelastic Scattering Cross Section

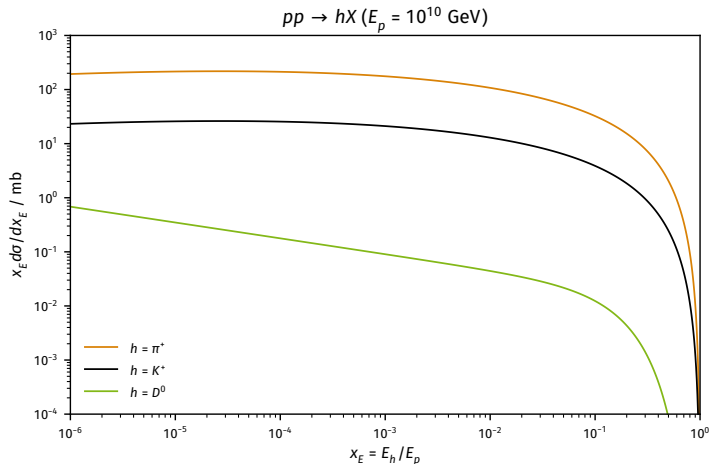


Combined Parametrizations

V. I. Belousov et al.
Phys. Atom. Nuclei (2016)

D. A. Fagundes, M. J. Menon
Nucl. Phys. A (2012)

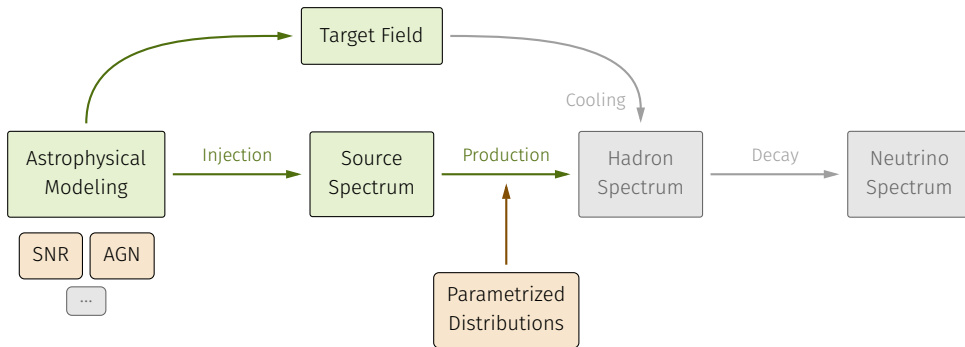
Weighted Energy Differential Cross Section

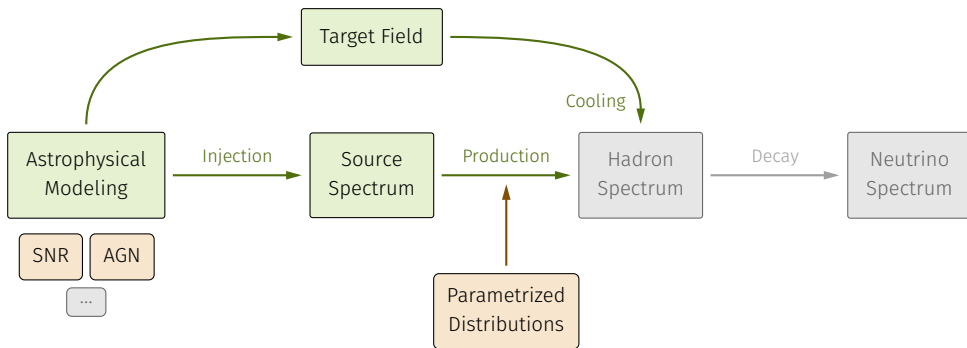


Derived Parametrizations

S. R. Kelner et al.
Phys. Rev. D (2006)

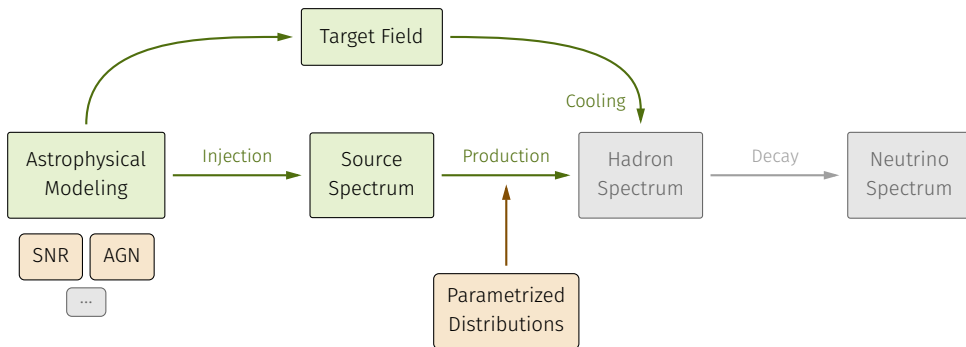
V. P. Gonçalves et al.
JHEP (2007)

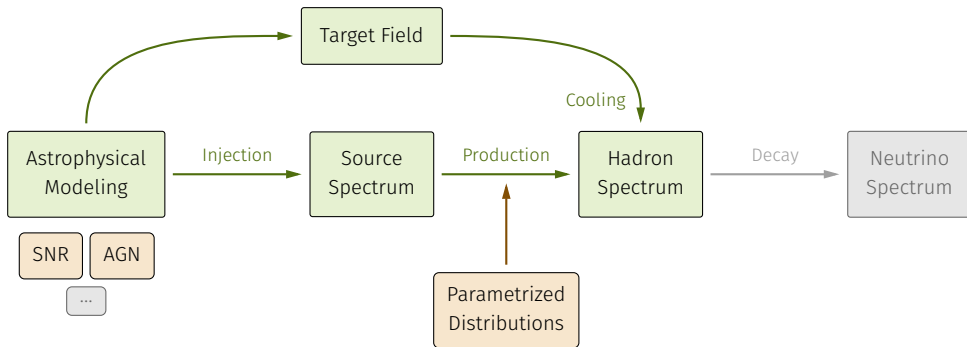




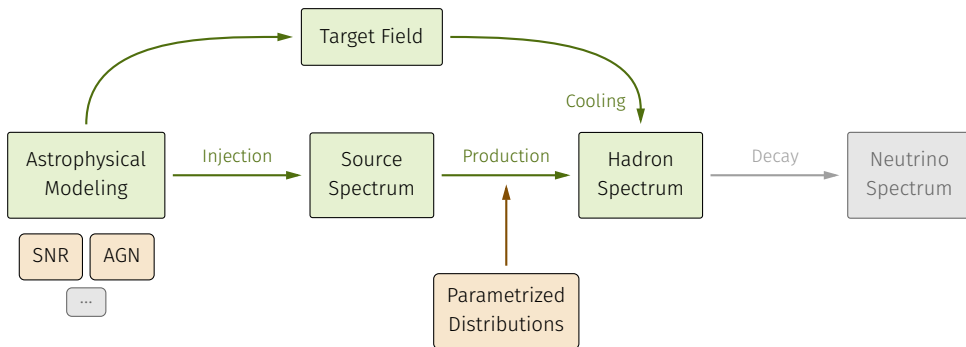


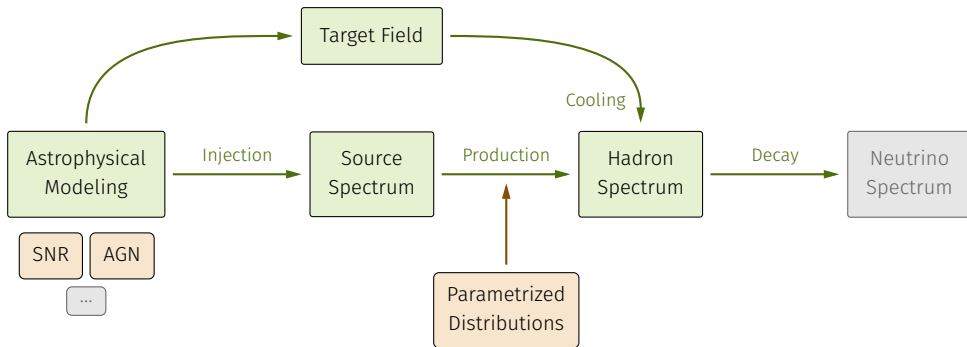
Cooling & Optical Depth



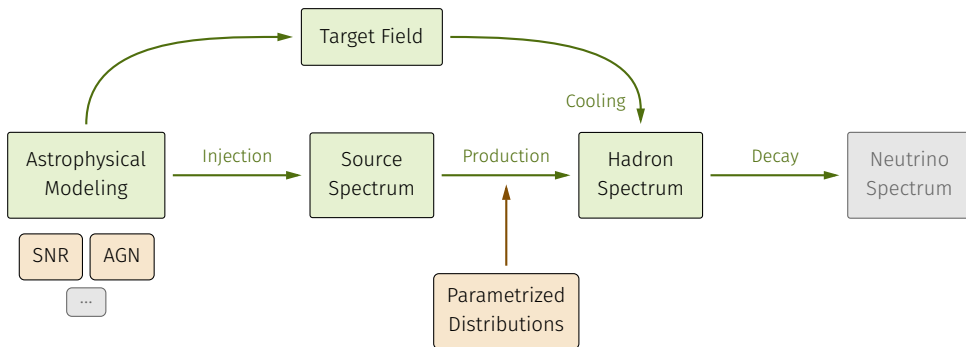


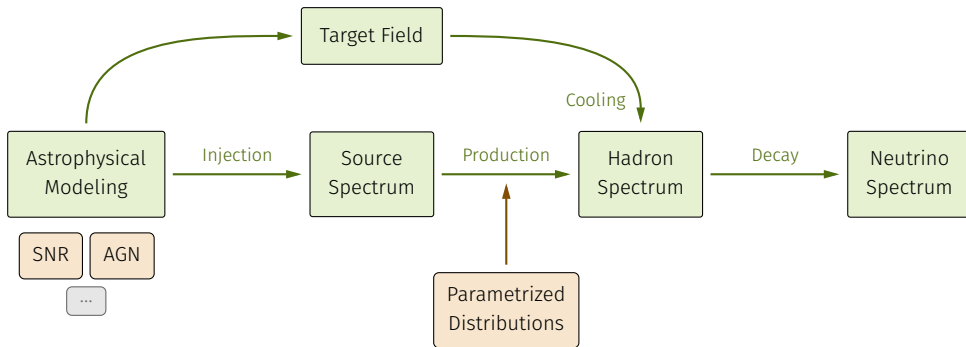




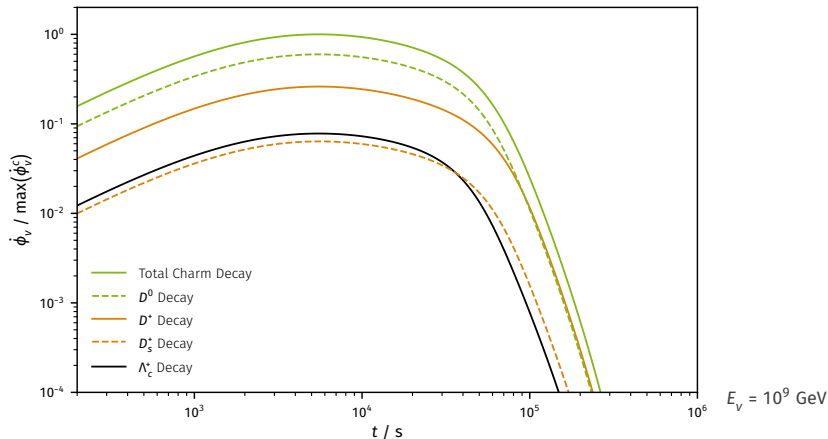




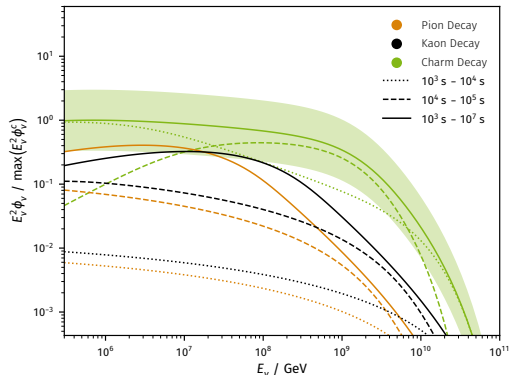
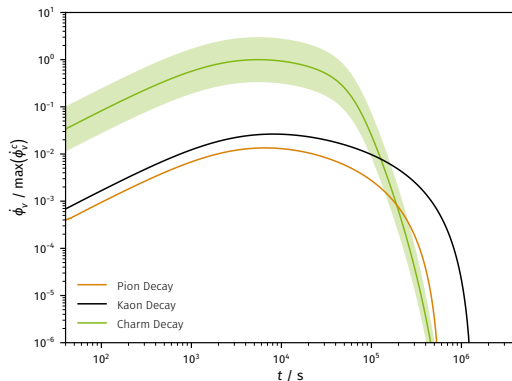




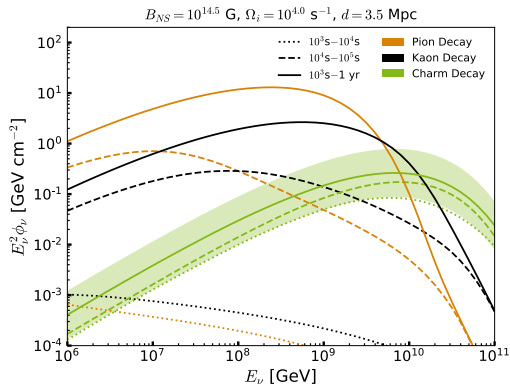
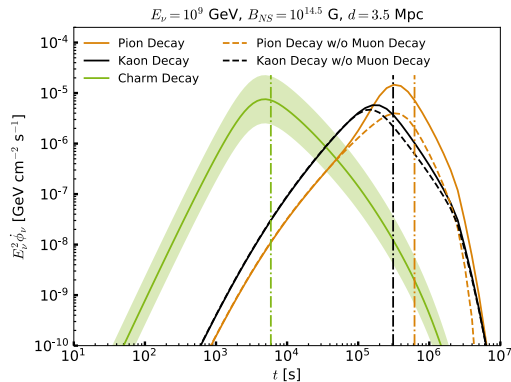
Normalized Magnetar Charm Contributions



Normalized Magnetar Neutrino Flux & Fluence

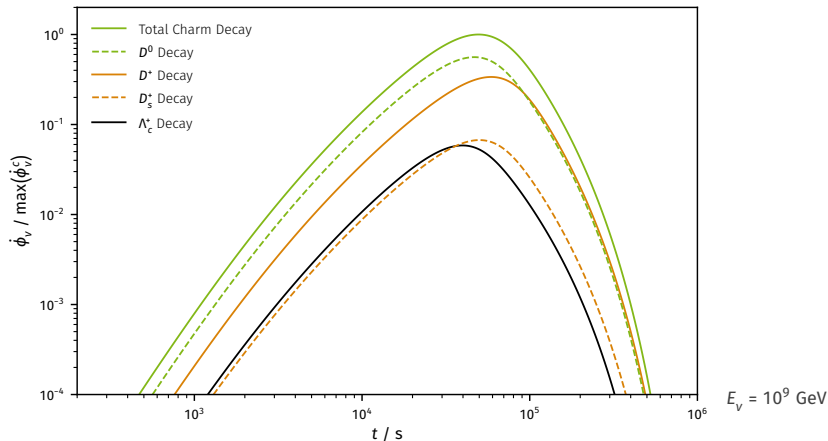


Published Estimated Magnetar Neutrinos

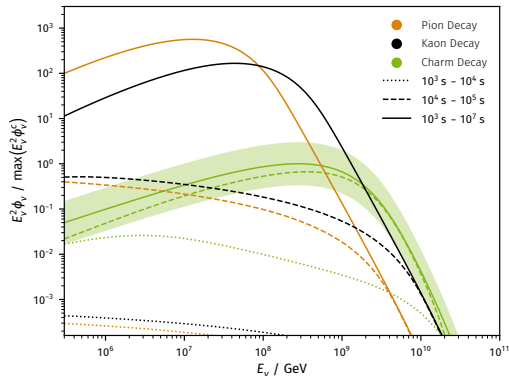
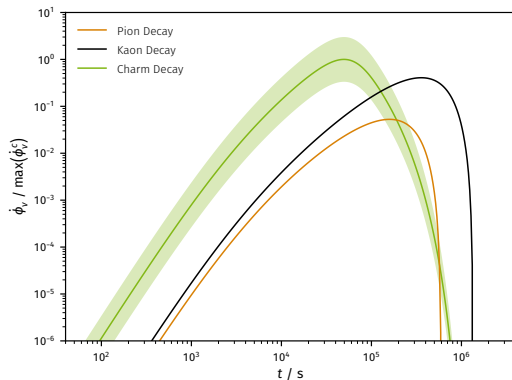


J. A. Carpio et al. *Phys. Rev. D* (2020)

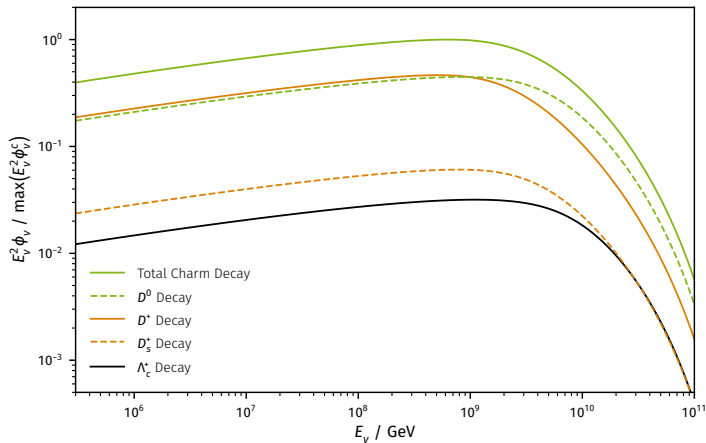
Modified Magnetar Charm Contributions



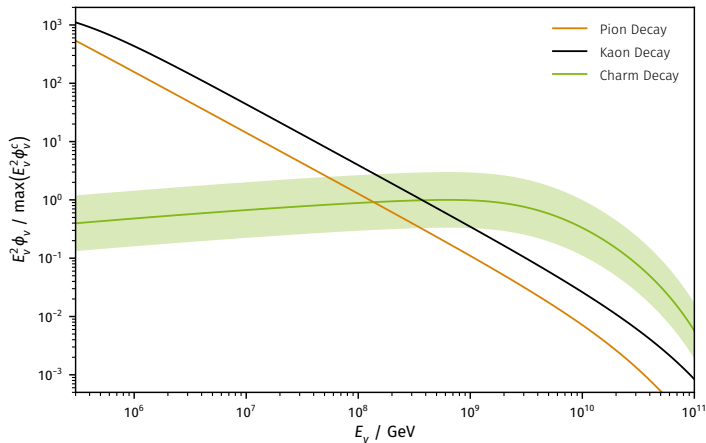
Modified Magnetar Neutrino Flux & Fluence

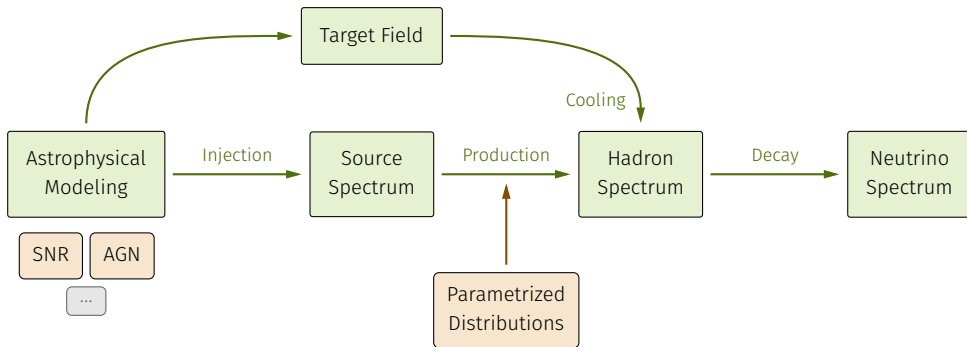


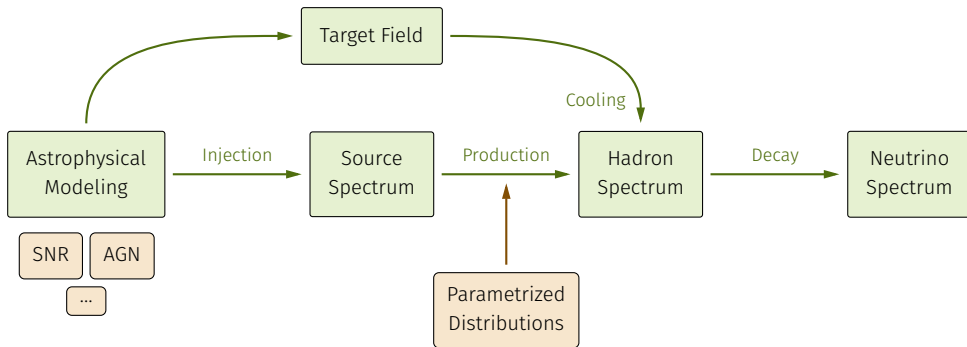
Normalized AGN Charm Contributions



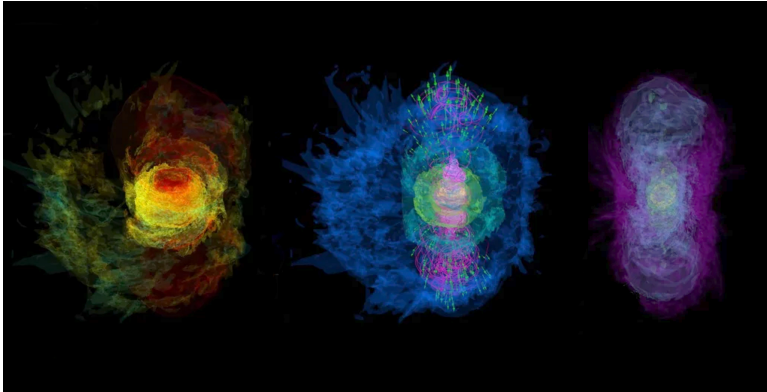
Normalized AGN Neutrino Fluence





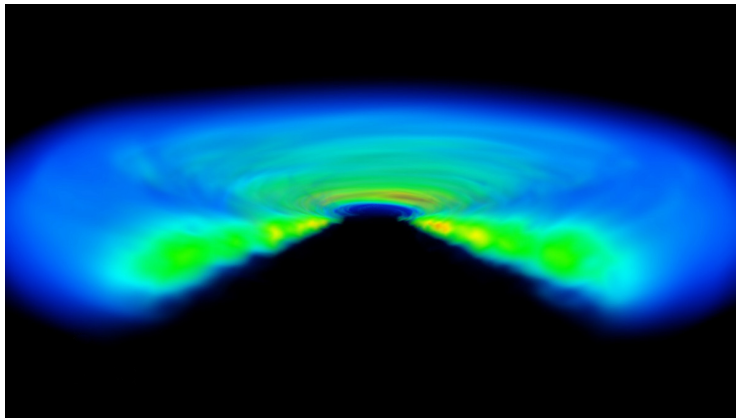


Simulated Magnetar Formation

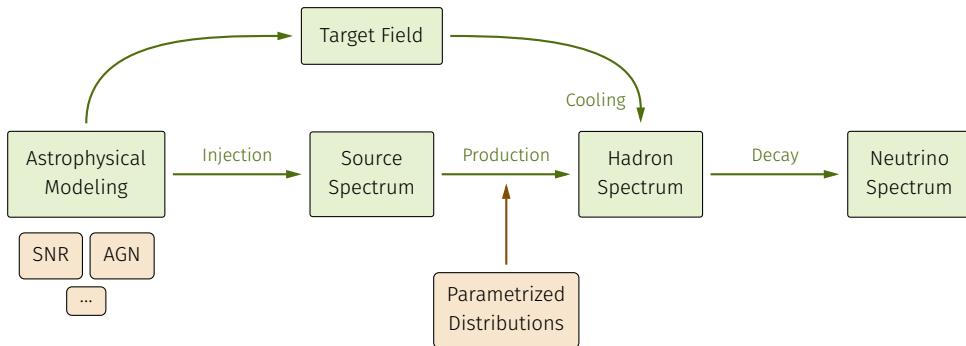


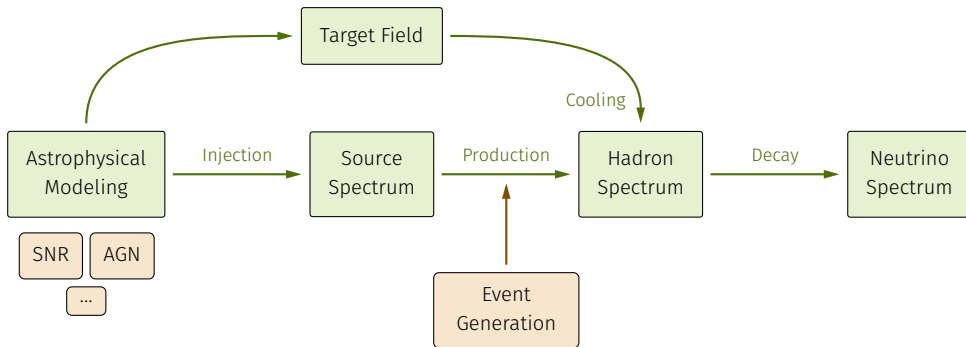
K. Hayashi *MPI Grav. Phys.* (2024)

Accretion Disk Structure

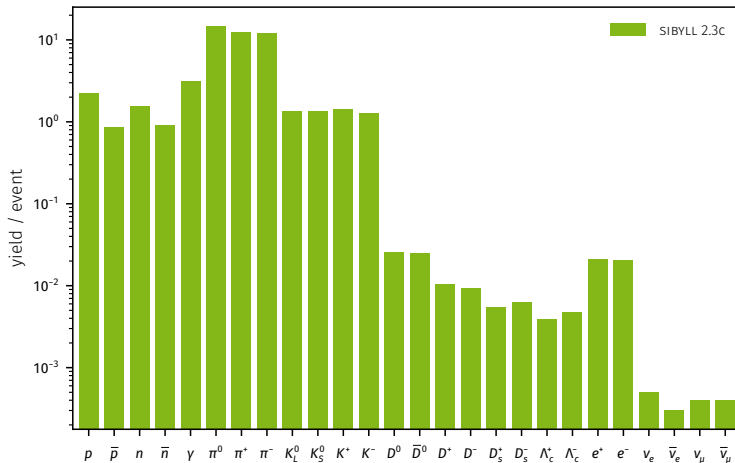


V. V. Zhuravlev et al. *ApJ* (2014)



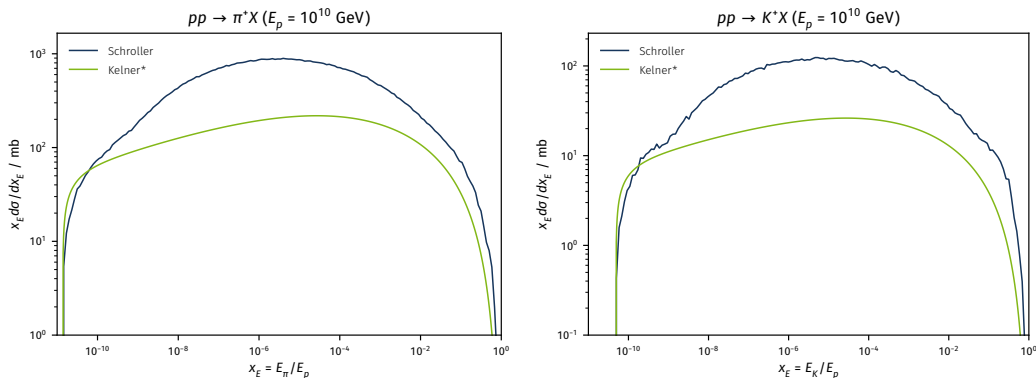


Monte Carlo Event Generation



Data Courtesy
M. Schroller

Differential Cross Sections Comparison



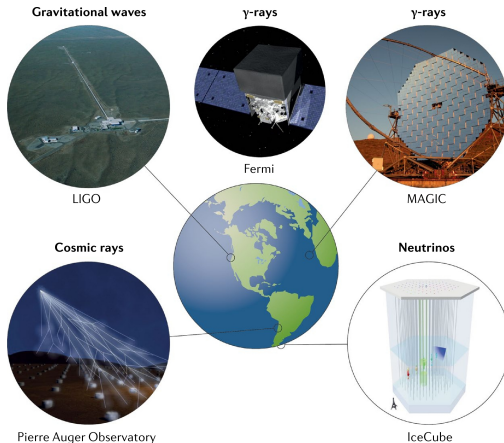
Derived Parametrization S. R. Kelner et al. *Phys. Rev. D* (2006)

Event Data (SIBYLL 2.3c) Courtesy M. Schrollier



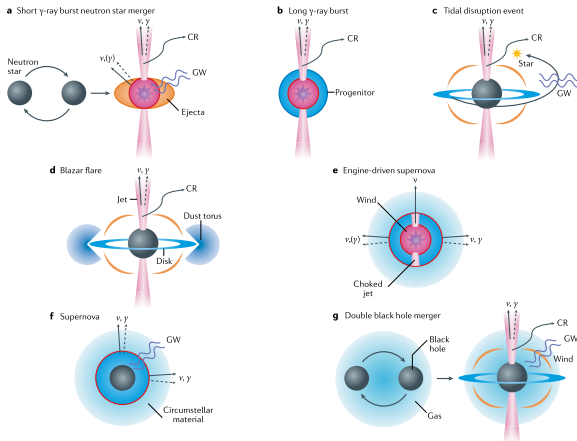
Summary & Conclusion

Messenger Types & Detection



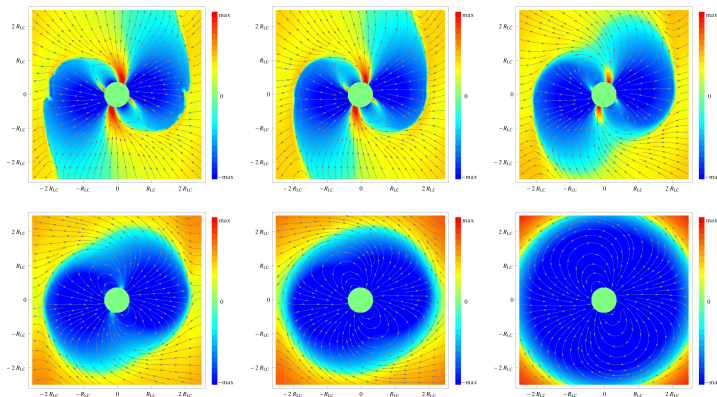
P. Mészáros et al.
Nat. Rev. Phys. (2019)

Source Regions & Events



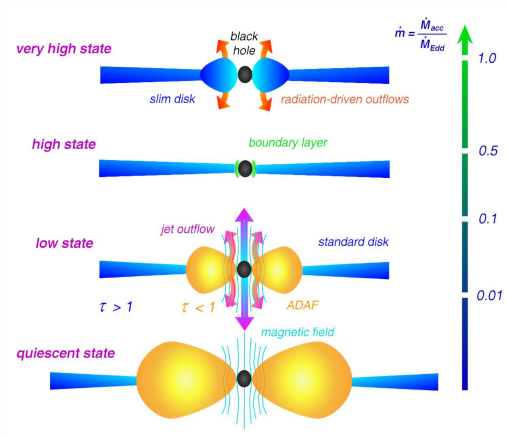
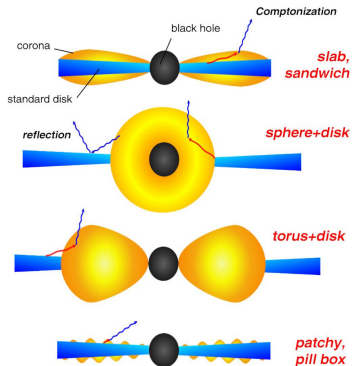
P. Mészáros et al.
Nat. Rev. Phys. (2019)

Resistive Pulsar Magnetospheres



J. Li et al. *ApJ* (2012)

Accretion Disk & Corona



A. Müller Heidelberg University (2004)